

Project Proposal

EN.705.641.81.SP26 - Natural Language Processing: Self-Supervised Models

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Problem

The existing solution for removing privileged documents preceding a legal case consists of individuals (humans) manually identifying which documents contain privileged information. The primary goal of this proposal will be on attorney-client privilege.

What is attorney-client privilege? It refers to confidential communication between a lawyer and a client (in this case a corporation) that are protected by law. These communications can include legal advice or sensitive company information. As such, these communications are not required to be shared with the opposing parties of the legal case. Since this requires individuals to read the contents to determine if it is privileged or protected, this task becomes very time-consuming and expensive for the firm. This process is far more complex than just searching for instances when a specific word appears or an email to an individual occurs. Not all communications between attorney and client are privileged. We aim to build a Natural Language Processing (NLP) model that can correctly identify privileged information and provide reasoning for why a document is classified as privileged or not.

Existing Approaches

*An Empirical Study on Transfer Learning for Privilege Review*⁷

This research study used BERT pretrained models to identify privileged documents within a dataset. The study made great strides when the test data domain was represented within the training dataset, but the model was unable to generalize to privileged documents in different domains. In their study, they indicated that the next steps would be to explore transformer models.

*Separating the Privileged Wheat from the Chaff – Using Text Analytics and Machine Learning to Protect Attorney-Client Privilege*⁵

This research study points out the fact that data nearly doubles every two years, which means that the manual review process for identifying privileged information would only become more expensive and require more people or more time (or both). This problem on top of rising costs means that the manual review process is unsustainable. At the time of their study, they claimed that “roughly 73% of all production costs occur during the document review process”. They claimed that the keyword searching within the documents is not sufficient for narrowing down the documents that must be reviewed. They implemented predictive modeling combined with keyword searching on a custom

set of keywords and showed that the combination of both practices resulted in the highest precision.

*The Future of Contract Review: Innovating Legal Practice through NLP Integration*³

This research study focuses on the fact that the review process for identifying privileged documents is prone to human error as well as very time-consuming. They stress the importance of adopting artificial intelligence, explicitly NLPs, to streamline the review process and increase the accuracy and consistency when identifying privileged documents.

*Snorkel: Rapid Training Data Creation with Weak Supervision*⁴

This paper provides a technique for creating a weakly labeled dataset using labeling functions. Creating a labeled dataset is very time consuming, so this solution mitigates the time needed to create a labeled dataset.

*Chain-of-Thought Prompting Elicits Reasoning in Large Language Models*⁶

This paper discusses how to implement Chain-of-Thought prompting, so the model provides a reasoning for each prediction. They show that only a few examples provided to the model can dramatically improve the model's performance.

Proposal

We propose to develop a Natural Language Processing (NLP) that will identify communication between attorneys and clients, and any other document considered privilege (e.g. Company secrets or Intellectual property (IP)). We will first test with the Enron Dataset but will expand to more datasets as needed if the project deems it necessary. We will base our approach relying on machine learning and NLP, which differs from traditional approaches which rely on keyword searches and metadata filtering to reduce the population of documents. Our approach will leverage semantic NLP so that our model can understand the context and intent of communications.

We will use the classic binary text classification model, where each document is classified as:

- Privilege
- Not Privilege

We will combine the proven techniques written in the papers we have referenced:

- Transformer Models (BERT). (Zhao 2021)
- Keyword and Machine Learning Hybrid (Keeling 2019)
- Weak Supervision (Ratner 2020)
- Instruction Tuning (Wei 2022)

This approach will be built upon prior research that demonstrates the effectiveness of transformers-based models for privilege classification. In addition, we will explore whether combining keyword-based methods with machine learning will improve the performance of the

model. Public datasets cannot contain privileged information, which is the reason why machine learning privilege review development has been limited. For this issue, we will use the technique discussed in the *Ratner 2020* journal “Weak Supervision”. Finally, we extend prior work by incorporating Instruction Tuning to enable the model to provide reasoning for its predictions, improving interpretability.

Data

There are not any public datasets that consist of privileged data since privileged data remains privileged indefinitely and thus cannot be made public. Privileged data often consists of emails between an attorney and a client. There is a famous dataset, the *Enron Email Dataset*², that consists of actual emails between employees at Enron. We propose to create a new dataset from the Enron dataset by using weak labeling with the following set of rules:

- Contains any of the following words: privilege, privileged, protected, legal, council
- Identifying sender/recipients that would be attorneys

Since not all communication between attorneys and clients is privileged, we would also instantiate noise within the weak-labeled privileged data to only include roughly 90-95% of the instances.

Methods

We plan to leverage the free GPUs on Google Colab to build and train classifications models that are fine-tuned using Instruction Tuning. The goal of the Instruction Tuning is to describe the reasoning behind why a document is identified as privileged or not. We will use pretrained models, like BERT and GPT, available publicly and then perform additional training on our dataset, followed by Instruction Tuning. The Instruction Tuning will require us to prepare examples of how we reasoned why an email/document within the dataset is or is not privileged.

Evaluation

Using standard metrics that are regularly used in the eDiscovery and prior research field for legal litigation we will evaluate the model:

- Precision: Here we will measure how many documents are actually labeled Privilege.
- Recall: This metric will be use to check how many true privilege documents are identified by our model successfully.
- F1 score: This metric will be used to measure the balance between precision and recall.

Recall is important, because privilege documents that are missing creates a big problem for a company where it could lead to an unwanted disclosure of confidential or privilege documents to the opposing party. On the other hand, precision is also important as it helps reduce the cost of manual review. Prior research in the field has shown that legal teams evaluate systems at a fix recall level of 75% and then measure precision at that level. (Zhao 2021)

In order to test these methods, we will follow the following workflow for a baseline:

- Keyword filtering
- Logistic Regression
- BERT (baseline)

Our Model on the other hand:

- BERT with fine tuning
 - o This will capture word semantics, making our model understand relationships between words.
- BERT with a hybrid model
 - o This will incorporate keyword signals and metadata as well as test whether combining traditional methods with semantic models will improve performance.
- BERT, Hybrid and Instruction Tunning
 - o Add reasoning capacity
 - o Generates explanations for classification decisions

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